

UNSUNG WOMEN WARRIORS OF INDIAN SCIENCE SERIES

THE TRAILBLAZING JOURNEY OF
**ROHINI
GODBOLE**



K. Venkataraman, Amritanshu Vajpayee,
Nandini Phanse and Shilpa Parikh

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ROHINI GODBOLE



Rohini Madhusudan Godbole (1952–2024)

National Pledge

India is my country.

All Indians are my brothers and sisters.

I love my country,
and I am proud of its rich and varied heritage.

I shall always strive to be worthy of it.

I shall give respect to my parents, teachers
and elders and treat everyone with courtesy.

To my country and my people, I pledge my
devotion.

In their well being and prosperity alone,
lies my happiness.

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THE TRAILBLAZING JOURNEY OF ROHINI GODBOLE

Authors: K. Venkataraman, Amritanshu Vajpayee,
Nandini Phanse, Shilpa Parikh

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DEDICATED TO ALL THOSE IN THE
PURSUIT OF KNOWLEDGE.

चरन्मार्गान्विजानाति ।

A WANDERER (EVENTUALLY)
FINDS THE PATH.

**उत्तिष्ठत जाग्रत प्राप्य वरान्निबोधत ।
क्षुरस्य धारा निशिता दुरत्यया दुर्गमपथः ॥**

"Arise, awake, and learn by
approaching the noble ones.
The razor's edge is sharp and
hard to cross
— so is the path to wisdom."

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Foreword

On the rich canvas of Indian science, few lives weave together the threads of intellectual rigor, unyielding perseverance, and quiet advocacy as beautifully as that of Professor Rohini Madhusudan Godbole. Born in the early years following Independence, she rose from the classrooms of a Marathi-medium school in Pune to the forefront of global particle physics, all while championing the cause of women in STEM with a voice that was both measured and resolute. This book captures not just her trailblazing journey, from developing early days at TIFR for the detection of the top quark, unravelling supersymmetry at IISc, to editing *Lilavati's Daughters*, but the essence of what it means to build science in a young nation: curiosity sustained by disciplined effort, collaboration, and a commitment to lifting others as one climbs.

Against the tides of time, her spirited character carried her to first rank at S.P. College, a silver medal at IIT Bombay, and a Ph.D. from Stony Brook in 1979, where she trained in quantum field theory and particle physics, laying the foundation for her later work on top quark phenomenology. Back home at TIFR, she didn't chase glory abroad; she instead chose to craft collider phenomenology that contributed to theoretical frameworks that were later reflected in the 1995 discovery of the top quark at Fermilab, then at IISc's Centre for High Energy Physics, contributing to studies of Higgs physics and supersymmetry relevant to future experiments such as the International Linear Collider.

What strikes the most about Rohini's story is its humanity. She faced the usual hurdles, resource constraints, long-distance choices, the relentless grind of calculations that don't always yield, but turned them into lessons for generations. Her Ph.D. at Stony Brook, her Padma Shri, her battles for gender equity weren't trophies; they were tools to open doors for students from Huzurpaga to Harvard. She mentored Ph.D.s, turning her cabin into a hive of questions and then there was her enduring legacy: calling out the "missing women" in physics not as a side project; instead, it was her moral compass, earning her the Padma Shri in 2019 and France's *Ordre National du Mérite* in 2021. Her real awards were the students who walked taller.

For young minds in Vidyarthi Vigyan Manthan or any classroom, this narrative is not simply a biography. It is a blueprint: ask sharper questions, persist through doubt, and remember science thrives when everyone gets a fair shot. Rohini Godbole didn't just predict particles, but rather she illuminated paths, she showed how curiosity, corrected by evidence and kindness, forges nations. May her light guide you forward. So read on, question boldly, and carry her fire.

Shekhar C. Mande, *FNA, FASc, FNASc*

National President, Vijnana Bharati

President, Indian National Science Academy (INSA)

Former Director General, CSIR

Distinguished Professor, Savitribai Phule Pune University and

JC Bose Fellow, National Centre for Cell Science, Pune

Preface

माणूस मोठा कर्मनि । नसतो तो जन्माने ॥
Greatness comes from one's deeds,
not from one's birth.

Life stories of scientists serve several purposes. They preserve individual achievement, record the growth of institutions, and help us see how knowledge advances through real human lives. For students, such narratives have a special value: they show that science is not created by mysterious “geniuses” alone, but by people who learn, struggle, make mistakes, seek help, and keep going.

The life of Professor Rohini Madhusudan Godbole (1952–2024) offers a particularly rich opportunity to understand these processes. Born in Pune less than a decade after India's Independence, she grew up in a household that respected books and learning even when the access to resources was quite limited generally. She started her journey from a Marathi-medium girls' school, then graduating from a Pune college went to Indian Institute of Technology Bombay in 1972 for her M.Sc. in Physics, and then, in 1974, to Stony Brook University in the United States for a Ph.D. in theoretical particle physics, which was completed in 1979. She went on to work at the Tata Institute of Fundamental Research (TIFR), the University of Mumbai, and finally the Indian Institute of Science (IISc) in Bengaluru, where she joined the Centre for Theoretical Studies (later the Centre for High Energy Physics) in November 1995 and remained associated until her passing in October 2024.

Scientifically, Professor Godbole was a leading figure in particle physics phenomenology. Her work on the structure of high-energy photons, on the production and properties of the top quark, and on Higgs and supersymmetric particles at present and future colliders helped shape how experiments were planned and interpreted. She was not only a contributor to the phenomenology of the Standard Model and physics beyond it, but also an enthusiastic champion of major international projects such as the

International Linear Collider. At the same time, she remained deeply engaged with teaching and supervision: she guided many Ph.D. students, several M.Phil. and Master's students, and continued working with students and collaborators even after formal retirement from IISc in 2018.

Equally important was another dimension of her work. Over the years, Professor Godbole became one of the most respected voices for women in science in India. She chaired the Indian Academy of Sciences' Panel for Women in Science, helped prepare influential reports on the loss of trained women scientists from the system, and co-edited two widely read books—*Lilavati's Daughters: The Women Scientists of India* (2008) and *The Girl's Guide to a Life in Science* (2011)—which have inspired many students to consider scientific careers seriously. In 2019 she received the Padma Shri for her contributions to Science and Technology, and in 2021 she was honoured by France with the *Ordre National du Mérite*, recognitions that reflected both her research and her advocacy.

This biography is written specially for students participating in National Talent Search Examination Vidyarthi Vigyan Manthan jointly organized by Viñāna Bhārati, NCERT and NCSM from Classes VI to XI in mind. Its language aims to be clear and engaging, without oversimplifying the science or romanticizing the difficulties. When technical ideas appear—such as the Standard Model, colliders, or supersymmetry—they are explained slowly and in context. The focus, however, remains on habits and values that sustain a life in science: curiosity, careful thinking, respect for evidence, perseverance in the face of setbacks, and a sense of responsibility towards others.

Teachers may find material here for classroom discussion not only in physics, but also in value education, social studies and career guidance. The chapters are interspersed with short reflection prompts and explanatory boxes. The narrative draws on multiple authentic sources: reminiscences written by colleagues, interviews given by Professor Godbole herself, institutional records, and published profiles and obituaries. Care has been taken to check factual details such as dates, institutional affiliations, and the sequence of events; where there are small differences between sources, the most reliable and recent accounts have been followed.

If, after reading these pages, even one student feels encouraged to ask a deeper question about nature, to read a little more carefully, or to imagine a place for themselves in the scientific community, then the purpose of this biography will have been served. And if some readers also feel moved to think about fairness, inclusion and the responsibility we bear towards our

institutions and our society, they will be walking in the spirit of Professor Rohini Godbole's own life.

K. Venkataraman
Amritanshu Vajpayee
Nandini Phanse
Shilpa Parikh

List of Abbreviations

AEPSHEP	Asia–Europe–Pacific Schools in High Energy Physics
APS	American Physical Society
B.Sc.	Bachelor of Science
CERN	Conseil Européen pour la Recherche Nucléaire (European Council for Nuclear Research)
CHEP	Centre for High Energy Physics
CSIR	Council of Scientific & Industrial Research
CTS	Center for Theoretical Studies
DST	Department of Science & Technology
IASc	Indian Academy of Sciences
IISc	Indian Institute of Science
IISER	Indian Institute of Science Education and Research
IIT	Indian Institutes of Technology
ILC	International Linear Collider
INSA	Indian National Science Academy
IUPAP	International Union of Pure and Applied Physics
LHC	Large Hadron Collider
LIGO	Laser Interferometer Gravitational-wave Observatory
M.Sc.	Master of Science
MSIL	Mysore Sales International Limited
NASA	National Aeronautics and Space Administration
NASI	National Academy of Sciences, India
NCERT	National Council of Educational Research and Training
NCSM	National Council of Science Museums
NSTS	National Science Talent Scholarship
Ph.D.	Doctor of Philosophy
QCD	Quantum chromodynamics
ROSCOSMOS	State Corporation for Space Activities “Roscosmos” commonly known simply as Roscosmos (Russian: Роскосмос)
S.P. College	Sir Parashurambhau College
STEM	Science, Technology, Engineering and Mathematics
SUNY	State University of New York
SUSY	Supersymmetry
TIFR	Tata Institute of Fundamental Research
WHEPP	Workshop on High Energy Physics Phenomenology

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CHAPTER - 01

A Girl Who Loved Questions

जिज्ञासा ज्याच्या अंतरी । त्यासी ज्ञान जवळी ॥

One who holds curiosity within the
heart, gains knowledge.

A beginning in an evolving modern nation

The decade after India became independent on August 15, 1947 was a time of rebuilding and planning. That period following India's Independence was marked by aspirations of shaping a future developed India. It was a tight rope walk for a nation in its infancy amidst a politically aligned bipolar split world. Institutions of higher learning were being strengthened, scientific laboratories were taking shape, and families across the country were beginning to see education as the surest path toward individual dignity and national development. It was the era of Indian scientists, engineers and technocrats like Sir C.V. Raman, Shanti Swaroop Bhatnagar, Prof. Satyendra Nath Bose, Dr. Meghnad Saha, Dr. Homi Jehangir Bhabha, Dr. Vikram Ambalal Sarabhai and other like-minded brains who had a self-guided roadmap supported by the newly independent Indian administrative, executive and governmental framework to lay the foundations of future developed India, were sketching ambitious futures for science in the country.

While Dr. Shanti Swaroop Bhatnagar was working tirelessly to establish a pan – India framework of CSIR which would serve as the artery system of future science research and administration, Dr. Homi Jehangir Bhabha was spending days and nights without fail to establish the modern Indian nation, still in infancy post – Independence, on the scientific and nuclear power map of the new global order. At the same time, Dr. Vikram Ambalal Sarabhai had a vision of an India that would be self-reliant in the field of communication network and space exploration. In that era, the United States and the Soviet Union were engaged in the space race, competing neck to neck

to win the race to send humans to space, Dr. Sarabhai had a greater vision, where he was working to utilize the space technology to support the agrarian economy and create a pan-Indian satellite communication network that would not only connect the Indians but also lead to help in security needs. Dr. Meghnad Saha – a pioneering astrophysicist was giving his contributions in making a self – reliant and scientifically vibrant India in his own way. The National Calendar of India that India adopted in 1956 which was built on the foundations of the amalgam of tradition Indian astronomical wisdom and modern astronomical principals in one such example. While academicians like Sir Chandrasekhara Venkata Raman, Prof. Satyendra Nath Bose were toiling hard to establish India on the global research map by developing infrastructure for Indian research institutions with the support of Indian and global scientific fraternity.

In this atmosphere of cautious hope and rebuilding, on 12 November 1952, a daughter was born in Pune to Madhusudan and Malati Godbole. They named her Rohini (IAST pronunciation: *Rohiṇī*), after a star (to be more accurate a *nakshatra*). No one could have predicted that she would



(Figure 1.1: Prof. Rohini Godbole with her sisters and nieces at a family function on 27 March 2024, Image courtesy: Sunil-Mukhi-Memories-of-Rohini-Godbole-2024-03-14-IISc, source: <https://rmgodbbole.in/>)

one day help place India firmly on the world map of particle physics, while also becoming a recognisable face of the movement for gender equity in science.

From the outside, the Godbole family looked like many other educated, middle-class households of that period. Theirs was a cultured and highly-educated middle-class family, where different members felt free to pursue different academic disciplines. Her father, Madhusudan Godbole, had been a topper in Economics at the University of Mumbai, but could not pursue academics further due to financial constraints. Instead, he joined the Defence Accounts, from where he retired in a senior position. Her mother, Malati, a high school topper, excelled in Sanskrit, but early marriage came in the way. Later she completed her Master's in Hindi and became a school teacher. Inside the home, two quiet forces were at work: unfinished dreams and firm resolve. Both parents had won scholarships and loved their subjects - Economics in the father's case, Sanskrit in the mother's - however, circumstances had cut short their own studies in their youth. Instead of turning that disappointment into bitterness, they converted it into determination: their four daughters would be allowed to explore the horizons of knowledge more than ever before. Rohini's siblings are all highly educated, Mrunalini (now Sunita Karlekar) studied and taught Mathematics, Kalpana (now Kelkar) became a medical doctor and a professor of Anaesthesiology, and the youngest Vibhawari (now Thakar) also worked as a science and mathematics teacher.

Books were treated with respect in the Godbole family. Time for homework was protected. Examinations were taken seriously, not because marks were everything, but because effort mattered. Above all, higher education was never presented as something “unnecessary” for girls. It was spoken of as a natural continuation of interest and ability. In such an environment, ambition did not need to be loudly declared; it could grow quietly and steadily.

A young Rohini grew up with her three sisters in an atmosphere of affection, conversation and what might be called “friendly competition”. In a disciplined environment space, books and money were shared. Opinions were discussed, sometimes argued and then reconsidered. Cooperation and individuality learnt to live side by side. In later years, colleagues observed that Professor Godbole had a deep respect for collaboration - a value whose early traces could be seen in the lively and reflective household shared with her sisters.



(Figure 1.2: A memorable moment from the film: “All That Matters” on Prof. Rohini Godbole with Rohini Godbole and her sisters seeing old family photographs. Image source: <https://rmgodbole.in/>)

Discovering science in a “Home Science” timetable

Rohini’s school, the well-known Huzurpaga Girls’ High School in Pune, had a proud history of educating girls. Yet like many institutions of its time, it reflected social expectations: up to Class VII, the formal timetable offered “Home Science”, not “General Science”. It would have been easy to accept this quietly and move on. Instead, a turning-point arrived in the form of an examination.

When she was in Class VII, Rohini decided to sit for the State Scholarship Examination, in which one paper was devoted to General Science. No girl from Huzurpaga had ever won this scholarship. The syllabus went far beyond what was taught in her classroom. On paper, the attempt did not look “practical”.

Her teachers, however, did not discourage her. They did something more demanding: they agreed to teach her science outside school hours and even on holidays. This commitment and decision, taken in a staff room and a few modest homes in Pune, helped shape the story of Indian physics and showed the young student that her curiosity was valued.

One of those adults was particularly important. Rohini’s mathematics teacher, Mrs Sowani, asked her to come to her house, because her husband, Bhau Sowani, was known as an excellent science teacher. Under his

guidance, preparation for an exam turned into a first real encounter with the world of science. He did not merely dictate definitions; he introduced her to the Marathi popular science magazine *Srishti Dnyan*, encouraged her to read outside the syllabus, and pushed her to write essays and take part in science competitions. Science, in his hands, was not just a collection of facts; it was a way of asking sharper questions about everyday phenomena.

Rohini worked hard. She read, solved problems, and learnt to enjoy the feeling of understanding something that had once seemed difficult. She appeared for the scholarship examination—and succeeded. Only a small number of such scholarships were awarded that year; she won one of them. The scholarship amount, she later recalled with amusement, was just one rupee less than her entire school fee, yet its real value lay in something else: it proved to her that she could compete in a larger academic world if she was willing to prepare for it.

At the same time Huzurpaga was focussing on overall personality development of students encouraging students to write and speak, and Rohini took full advantage of this. She wrote short Marathi pieces for the school magazine *Balikadarsh*, including essays on historical personalities and on science, and took part in debates and elocution competitions. Standing in front of classmates and teachers, trying to organise her thoughts and speak clearly, she began to develop the confidence that would later make her an outstanding communicator of physics to students and the general public.

From S.P. College to a national stage

After school, Rohini joined Sir Parashurambhau College (S.P. College), affiliated to the University of Pune (now Savitribai Phule Pune University-SPPU), to pursue a B.Sc. in Physics, Mathematics and Statistics. Here again she worked with steady seriousness. By the time she graduated, she had stood second in the Intermediate examination and first in the B.Sc. in the university. This first rank brought with it the National Science Talent Scholarship (NSTS), a centrally administered scholarship for promising science students.

At home, this scholarship changed the conversation. With four daughters and a single government salary, it was not easy for her father to consider sending one child to a residential institute in another city. The NSTS amount—about Rs 200–250 per month at that time—was, as she later said, “a princely sum” for a middle-class family. It made it possible for her to think seriously of joining an Indian Institute of Technology for

Roots of Learning in Pune's Historic Institutions

Rohini Godbole's educational journey from the pioneering classrooms of Huzurpaga Girls High School to the scholarly halls of Sir Parashurambhau College (S.P. College), is a classic example of Pune's educational legacy. Many students, especially girls, followed this path. Both Huzurpaga Girls High School and S.P. College occupy a special place in the educational history of Maharashtra, particularly in advancing women's education and access to higher studies.

Founded in 1884 in Pune, in the pre-independence era, **Huzurpaga Girls School**, was one of the earliest schools for girls in India. At a time when girls' education was discouraged, eminent social reformers like Justice Mahadev Govind, R. G. Bhandarkar, Shankar Pandurang Pandit and Waman Abaji Modak felt the need to educate women and founded the school. Huzurpaga Girls High School became one of the earliest institutions to provide structured, modern schooling to young women. It helped cultivate literacy, confidence, and intellectual curiosity among girls who otherwise had little access to formal education. Many of its students went on to become teachers, reformers, doctors, writers, and scientists, demonstrating the transformative power of girls' schooling.

Sir Parashurambhau College, established in 1916, in Pune, under the aegis of the Shikshana Prasarak Mandali, opened higher education opportunities for women. Named after social reformer Sir Parashurambhau Patwardhan, it offers opportunities in Arts, Science and Commerce streams, enabling women to pursue advanced learning, research, and professional careers.

Together, these institutions created an educational pathway—from school to college—that allowed generations of women to grow into independent thinkers and contributors to society. These historic institutions continue to inspire students to pursue knowledge with courage, creativity, and social responsibility.

Because of many such institutions, women today stand as teachers, scientists, administrators, and innovators. They remind us that when we educate a girl, we educate a generation. Their legacy continues the vision of pioneers like Savitribai Phule, who believed that educating girls was the key to social progress.

her M.Sc. Without that financial support, she believed, IIT might have remained out of reach.

After her graduation, Rohini knew that physics was the subject that drew her most strongly. She also enjoyed mathematics, which provided precision and structure, while physics applied this mathematical language to explain natural phenomena, from planetary motion to the behaviour of invisible particles.

In 1972, after completing her B.Sc., she applied for admission to the M.Sc. Physics programmes at both IIT Bombay and IIT Kanpur, then considered two of the best destinations in the country for advanced physics. At that time, IIT Kanpur had a slightly stronger reputation in physics, and naturally it attracted her interest. Her parents, however, reminded her of a practical fact: if she studied in Bombay and fell ill or faced difficulty, they could reach her in about four hours; if she studied in Kanpur, it would take nearly nineteen hours to get there. After weighing these points, Rohini decided to join IIT Bombay for her M.Sc. in Physics.

This decision—rooted in a mixture of ambition, family concern and simple geography—shaped the rest of her life. At IIT Bombay, she graduated at the top of her class and won the Institute Silver Medal in 1974. This was also the place where she met Professor S. H. Patil who helped her discover the beauty of theoretical physics, and decide to become a theorist. That part of the story shall unfold in the next chapter.

Think Like a Physicist- Try to Imagine

When encouragement arrives at the moment of uncertainty, it can redirect an entire life. Reflect on how intellectual trust — given or received — shapes the courage to attempt difficult tasks.

Student Reflection

Major choices often require both personal effort and collective support. Consider how families, teachers, and institutions can create cooperative environments in which talent blooms.

Chapter - 02

Learning to Become a Physicist

विद्येची साधना हीच खरी प्रगतीची साधना.

The pursuit of knowledge is the true path to progress.

By the early 1970s, the foundations of modern particle physics were being laid in laboratories and universities across the world. The ideas that would later form the Standard Model were being refined, and new experiments were being planned to test them. In 1972, at this very moment in scientific history, Rohini Godbole stepped onto a national stage as an M.Sc. student in Physics at the Indian Institute of Technology Bombay (IIT Bombay).



(Figure 2.1: Physicist Rohini Godbole. Her achievements compelled male scientists to count the missing women in their midst. | Photo Credit: N. Ashwini/The Hindu)

She arrived with a strong academic record from Pune University and the support of the National Science Talent Scholarship, but also with doubts that many students from regional-language backgrounds still recognise: Was her English good enough? Would she be able to keep up with the pace of teaching in such a competitive institution?

Do You Know - What is Particle Physics?

Particle physics attempts to identify the most fundamental building blocks of matter and the forces through which they interact. Instead of studying everyday objects, scientists investigate entities far smaller than atoms. Because such particles cannot be seen directly, researchers rely on mathematical models and highly sophisticated detectors. The aim is to discover simple principles capable of explaining enormous variety.

IIT Bombay: adjusting to a new rhythm

IIT Bombay in the early 1970s was already a demanding academic environment. For Rohini, it meant challenges on several fronts all together — new city, new hostel, new teaching style and a new language of instruction. She joined the M.Sc. Physics course, in a class of about seventeen students, only three of whom were women.

Her earlier education had been primarily in Marathi, and although she had studied English as a subject, she had never before studied physics *through* English. She later remarked that her biggest anxiety in those days was not about being a woman in a science classroom, but about expressing herself in English. She worried about whether she would understand everything clearly and whether she would be able to ask and answer questions in class with confidence.

The academic culture was different too. Instead of only one big examination at the end of the year, there were regular quizzes, assignments, mid-semester tests and even open-book examinations. Students were expected to think independently, attempt problems not seen before and discuss ideas with classmates after class. This rhythm required a steady pace of work rather than last-minute preparation.

Rohini Godbole responded in the way she had always done: by working methodically. She read the recommended texts, asked questions when needed,

and learnt to treat English not as a barrier but as a skill that could be improved with practice. Over time, she discovered that the logic of physics could give her confidence even when the language felt unfamiliar.

Her M.Sc. class, though small and mostly male, turned out to be a supportive group. Respect in the classroom depended more on preparation

Do you know: What is Theoretical Physics?

Theoretical physics is the branch of physics that uses mathematics, logic, and models to understand the fundamental laws of the universe—without necessarily performing experiments directly. Theoretical physics seeks universal laws through mathematical reasoning and conceptual models.

Main Areas of Theoretical Physics:

- Quantum Mechanics – Behaviour of tiny particles
- Relativity – Space, time, gravity
- Cosmology – Origin of the universe
- Particle Physics – Fundamental particles
- Statistical Mechanics – Large systems & thermodynamics

and performance than on whether a student was male or female. This did not mean that gender bias did not exist in society, but it did mean that in this particular classroom at this particular time, she did not feel actively discouraged as a woman.

Hostel life in the Ladies' Hostel (LH) added another layer of experience. Sharing rooms and responsibilities with other young women from different parts of India, she made friends, went for lakeside walks on the campus, watched films in the Convocation Hall and took part in the cultural life of IIT Bombay. Between assignments and exams, there were moments of laughter, music and informal debates about everything from politics to poetry. It was a reminder that even in intense academic environments, human connections sustain effort.

Meeting a mentor: Professor S. H. Patil

Every scientist can usually name one or two teachers who changed the direction of their life. For Rohini, one such person was Professor S. H. Patil, a theoretical physicist in the Department of Physics at IIT Bombay.

During her M.Sc., she undertook a “home paper” with him—a guided project that required her to read beyond the standard syllabus, work through derivations on her own and then discuss them in detail. This was not easy work. Professor Patil had high standards, and he expected his students to justify each step of an argument, not just quote results. But as Rohini struggled and persisted, something important happened: she began to see the inner beauty of theoretical physics.

Until then, she had liked physics as a subject and had been good at solving problems set in textbooks and exams. Under Professor Patil’s supervision, she saw that equations could do much more than solve familiar exercises. Properly arranged, they could predict new phenomena, connect apparently unrelated facts and even suggest what experiments should be done next. Theory, she realised, was not just about calculations; it was an attempt to understand the deep structure of nature.

By the time she completed her project, she knew that she wanted to be a theoretical physicist. The idea of working at the frontier of thought—even if it meant years of hard effort with no guarantee of immediate recognition—appealed to her temperament.

Later, when she told Professor Patil that she wished to go abroad for a Ph.D. in theoretical particle physics, his first reaction was cautious. For one of her male juniors he had said, “Why stay in India? You should go outside.” To her, he initially asked, “Why do you want to go abroad?” The difference may have had many reasons—concern for a young woman travelling alone, or differing career timings—but Rohini, by then, had clarity. She replied that she had thought seriously about it and had made up her mind.

Once he understood her determination, Professor Patil gave her his full support and wrote a strong recommendation letter. That letter, along with her excellent academic record and scholarship, would help open the doors of one of the world’s leading centres in high-energy physics.

Standing first, stepping out

In 1974, Rohini completed her M.Sc. in Physics from IIT Bombay, standing first in her batch and receiving the Institute Silver Medal. This recognition mattered not only as an award, but also as a signal that her decision to aim high had been realistic. She had proved, to herself and to others that a student educated in Marathi-medium schools could excel in one of India’s most competitive institutes of science and technology.

Do You Know?

Quantum Field Theory

Imagine that empty space is not really empty. Instead, think of it like a vast, invisible ocean. What we call particles, like electrons, are actually tiny ripples or vibrations in this ocean. The ocean itself is the “field,” and each ripple is a particle. This idea helps scientists understand how particles are created, interact, and disappear.

In simple academic terms: Quantum field theory is the framework in which particles are understood as excitations of underlying fields that fill all of space.

Quantum Chromodynamics (QCD)

This is the part of physics that deals with a special kind of glue—the strongest force in nature. It binds tiny particles called quarks inside protons and neutrons. Through a related effect, it also holds the centres of atoms (nuclei) together. The name comes from a Greek word for colour, because the force works through a property called “colour charge,” which combines in such a way that particles remain bound.

In simple academic terms: Quantum chromodynamics is the theory of the strong interaction, describing how quarks and gluons interact to form protons, neutrons, and other Hadrons.

Electroweak Theory

For a long time, scientists thought that electricity and the weak nuclear interaction (responsible for certain types of radioactivity) were completely different forces. Electroweak theory showed that they are actually two aspects of a single force. It is like discovering that ice and steam are both forms of the same substance. At very high energies, these forces merge into one.

In simple academic terms: Electroweak interaction unifies the electromagnetic and weak forces into a single interaction.

The Standard Model

Think of the Standard Model as a kind of family album for the universe. Scientists have carefully identified the fundamental particles and arranged

them into a clear picture. This “album” includes quarks, leptons, and the particles that carry forces. It shows how everything around us is built from a small set of basic ingredients.

In simple academic terms: Standard Model of particle physics is the theory that classifies all known elementary particles and describes three of the four fundamental forces of nature (excluding gravity). It has been extraordinarily successful in predicting experimental outcomes. However, it does not answer every question. Mysteries such as dark matter, neutrino masses, and gravity’s integration remain open, motivating continued exploration.

Yet instead of treating this medal as a destination, she saw it as a beginning. If she wanted to contribute to particle physics at the international level, she needed deeper training. At that time, many of the most active groups in theoretical high-energy physics were in Europe and North America. She decided to apply abroad for a Ph.D.

Among the offers she received, one stood out: a place in the Ph.D. programme in theoretical particle physics at the State University of New York at Stony Brook (SUNY Stony Brook), in the United States. Stony Brook was then a major centre for particle theory, with close connections to some of the most important developments in the field. Joining it meant entering a wider conversation—a world of seminars, visiting scientists and intense discussions that would prepare her to contribute at the frontiers of knowledge.

Leaving for the United States in 1974 was a big step. For her family, it meant sending a daughter far away at a time when international travel was rare and expensive. For Rohini, it meant adjusting to a new country, culture and climate, and once again facing the challenge of language and self-confidence, this time on an even larger stage. But she had already crossed such transitions once, moving from Pune to IIT Bombay. Now, with the support of her family and teachers, she was ready to cross another.

Stony Brook: learning to think as a researcher

At Stony Brook, Rohini entered a community that lived and breathed particle physics. Lectures and seminars were filled with discussions on quantum field theory, quantum chromodynamics (QCD), electroweak theory and

the rapidly consolidating Standard Model of particle physics. Students and faculty argued over blackboards, scribbling equations, erasing them and writing them again as they tried to capture nature in mathematical form.

Rohini worked under the supervision of Professor Jack Smith, himself a student of Nobel laureate Peter Higgs and an early expert in QCD, the theory of strong interactions between quarks and gluons. Under his guidance, she learnt how to move from broad physical ideas to precise calculations that could be compared with experimental data. She did not restrict herself only to strong interactions; even as a student she worked on problems involving electroweak interactions, gradually building a broad understanding of the entire Standard Model.

The daily routine of a Ph.D. student in theoretical physics is laborious and not glamorous. It involves long hours at a desk or board, checking calculations line by line, reading research papers, attending seminars, and



(Figure 2.2: Stony Brook Physics Department Group Photo from 1977. Rohini Godbole is at front centre.)

discussing problems with supervisors and peers. For Rohini, these years were a training for patience and precision. She learnt that in high-level research, progress often comes after repeated failure, and that being “wrong” is not shameful if one is willing to correct oneself.

In 1979, she completed her Ph.D. in theoretical particle physics from Stony Brook. Her thesis work placed her firmly within the international community of particle physicists. She had now spent five years in an advanced research environment, learning how major questions in physics were framed, debated and gradually answered.

The decision to come home

At this stage in many scientific careers, the usual path is to move from a Ph.D. to one or more postdoctoral positions in Europe or the United States, building further experience and publications before taking up a permanent



Image courtesy: Sunil Mukhi, source: <https://bhavana.org.in/rohini-godbole-a-passion-for-science/>

job. Rohini had such opportunities. Professionally, it might have been easier to remain abroad for some more years, taking advantage of well-funded laboratories and dense research networks.

Yet during these years she felt a strong pull towards India. She knew that Indian institutions faced constraints—limited funding, fewer large experimental facilities, heavier teaching loads—but she also knew that a great deal could be built and strengthened at home. Returning to India would mean more challenges in some ways, but it would also mean a chance to help shape the growing field of high-energy physics in her own country, and to be closer to her family.

Do You Know – What are Theory and Phenomenology?

Theoretical physicists construct models using mathematics. Phenomenologists take these models and ask how their consequences might appear in real experiments. They help bridge imagination and measurement. Professor Godbole worked extensively in this area, ensuring that theoretical possibilities could be tested.

After careful thought, she took a decision that would define the character of her career: she chose to return to India soon after her Ph.D., in 1979, instead of staying abroad for many years. It was not the “usual” choice for an ambitious young particle theorist at that time, but it was entirely in line with the values on which she had been raised—commitment, responsibility and a belief that knowledge should circulate back to the society from which it arises.

Two Great Institutions, One Inspiring Journey

Rohini Godbole’s academic trajectory from IIT- Bombay to Stony Brook University, New York reflects a deep commitment to physics and to the global advancement of knowledge.

Indian Institute of Technology Bombay (IIT Bombay) is one of India’s premier institutes of engineering, science, and technology, located in Powai, Mumbai. Established in 1958, IIT Bombay is part of the prestigious IIT system and is internationally recognized for excellence in teaching, research, and innovation. The institute offers undergraduate, postgraduate, and doctoral programs in engineering, science, design,

management and humanities. With world-class faculty, advanced laboratories, and a vibrant academic culture, IIT Bombay has produced leaders in technology, entrepreneurship, academia, and industry across the globe.

The State University of New York at Stony Brook, commonly known as Stony Brook University, is a leading public research university located in Stony Brook, New York, USA. Established in 1957, as a college for the preparation of secondary school teachers of mathematics and science, it is part of the State University of New York (SUNY) system and is internationally recognized for excellence in engineering, computer science, physics, biomedical sciences, and interdisciplinary research. With world-class faculty, advanced laboratories, and strong collaborations, Stony Brook University plays a significant role in scientific discovery and innovation.

Her next destination was an institution that was already a major centre for research in fundamental physics: the Tata Institute of Fundamental Research (TIFR) in Mumbai. There, in the Theory Group, she began to find her own voice as a researcher, work on problems connected to the then-unknown top quark, and help build a culture of collider phenomenology in India. How that unfolded is the story of the next chapter.

Think Like a Physicist- Try to Imagine

If understanding nature requires years of training, how should one treat daily study? Is it routine — or preparation for discovery?

Student Reflection

What motivates a person to return and build institutions where difficulties are greater? How do ideas of responsibility shape professional choices?

Chapter - 03

Building Science at Home: TIFR and the University of Bombay (1979–1995)

ज्ञान, विज्ञान, श्रम, साहस । या चार दीपांनी देश उजळू ॥
विज्ञानाच्या शक्तीने सारे । नवे सुवर्ण भविष्य घडवू ॥

Knowledge, science, hard work, and courage, May
these four lights illuminate our nation.
Through the power of science, Let us build a bright
and golden future.

When Dr. Rohini Godbole stepped off the aircraft in India in 1979, doctorate freshly completed from Stony Brook, she was returning to a scientific landscape very different from the one she had just left. In the United States, she had been surrounded by well-equipped laboratories, a dense network of theorists and experimentalists, and a steady stream of international visitors. In India, resources were more modest, but there was something else she valued: the chance to help build a research culture in her own country.

Her first formal position was as a Visiting Fellow in the Theory Group at the Tata Institute of Fundamental Research (TIFR), Mumbai, beginning in 1979. There, over the next few years, she would start to develop the scientific identity for which she later became known: a particle physics phenomenologist, someone who connects deep theoretical ideas to concrete predictions that experiments can test.

TIFR: entering Indian high-energy physics

At TIFR, Rohini joined a small but active community of theoretical physicists working on high-energy physics. Senior scientists like K. V. L. Sarma, D. P.

Roy, Probir Roy and Biswarup Banerjee were already thinking about how the newly central Standard Model could be tested at existing and future particle colliders. For a young scientist fresh from Stony Brook, this was an excellent environment: demanding, but full of opportunity.

Do You Know - What Does a Collider Do?

A particle collider accelerates beams of particles to very high energies and brings them into collision. These impacts recreate conditions similar to those that existed shortly after the Big Bang. By analysing the debris, scientists infer the presence of short-lived or previously unknown particles. In this way, colliders function as microscopes probing extremely small distances.

One of the most important questions in particle physics at that time was the existence of the top quark. The Standard Model predicted a third-generation partner to the bottom quark, but it had not yet been observed experimentally. Physicists knew it must be heavy, but its exact mass and the best strategies to find it were still uncertain.

Rohini joined a collaboration with D. P. Roy and Sandip Pakvasa (visiting from abroad) to ask a very concrete question: *If top quarks are produced at high-energy colliders, what would their presence look like in a detector?* The group suggested that one should look for events in which there are hadronic jets (sprays of particles produced when quarks hadronise) accompanied by isolated high-momentum leptons (electrons or muons that stand out clearly from the surrounding activity). This combination, they argued, would be a characteristic signature of top quark production and decay.

The term *hadronic* derives from ‘hadron’, a class of particles composed of quarks bound together by the strong nuclear force. Protons, neutrons, pions, and kaons are all examples of hadrons. In contrast, particles such as electrons belong to a different class called ‘leptons’. Unlike hadrons, leptons are not made of quarks and do not participate in the strong interaction. This fundamental distinction shapes the behaviour of matter: hadrons form atomic nuclei under the influence of the strong force, while leptons interact primarily through the electromagnetic and weak forces.

Do You Know - What are quarks?

A quark is a fundamental particle and one of the basic building blocks of matter. Quarks combine in groups to form composite particles called Hadrons, the most familiar of which are protons and neutrons. These particles, in turn, make up the nuclei of atoms. They are elementary particles, meaning they cannot be divided into smaller components, and they interact through the strong nuclear force, which is carried by particles called gluons. Quarks come in six types: up, down, charm, strange, top, and bottom—but only the up and down quarks are found in ordinary matter. For example, a proton is made of two up quarks and one down quark, while a neutron consists of one up quark and two down quarks. Quarks are never observed alone because they are permanently confined inside larger particles. Understanding quarks is essential in particle physics, as they help scientists explain the structure of matter and the fundamental forces of nature.

The term «quark» was coined in 1963 by the physicist Murray Gell-Mann during his attempt to describe the fundamental building blocks of matter. In search of a name that was both distinctive and memorable, Gell-Mann had been turning over a word that sounded like "kwawk" in his mind, intending it to rhyme with "cork." Before settling on a spelling, he happened upon a curious line from James Joyce's *Finnegans Wake*: "Three quarks for Muster Mark!" The literary reference provided him with the perfect orthography for the sound he already had, and thus the term "quark" was formally adopted into physics. A subtle irony remains, however, as Joyce's line seems to demand a pronunciation closer to "mark," creating a small linguistic rift between the scientist and the poet. This blend of physics and literature has since become one of the most charming anecdotes in modern science.

Years later, in 1995 (officially), when the top quark was finally discovered at Fermilab in the United States, it was indeed through channels that involved patterns very similar to those Rohini and her collaborators had analysed earlier. This work made her one of the pioneers of collider-based phenomenology in India, showing how theory could guide experimental searches in a practical, detailed way.

Daily life at TIFR was intense but not glamorous. Calculations were done by hand and with simple computing facilities. Preprints from abroad arrived by post, sometimes weeks late. Yet there was a strong sense of purpose. The theorists knew that if they worked carefully, their ideas could influence how major experiments around the world were designed and interpreted.

Science and people: mentorship and collaboration in Mumbai

TIFR was also where Rohini began forming collaborations that would last for decades. She worked with senior colleagues, but also with younger researchers and students. One of them was Sourendu Gupta, then a graduate student who would later become a recognised expert in nonperturbative QCD. Their early joint work gave her experience in team science, where different members bring complementary skills.



(Figure 3.1: During the visit of Manuel Drees to TIFR in January 1993 (L to R: Dileep Jatkar, Ashoke Sen, Manuel Drees, Sumathi Rao, and Rohini Godbole). Image courtesy: Manuel Drees, source: <https://bhavana.org.in/rohini-godbole-a-passion-for-science/>)

Personality mattered in these interactions. Rohini's colleagues recall that she brought a mixture of seriousness and warmth to discussions. She was not shy about criticising an argument if the physics was weak, but she did so in a way that focused on the idea, not on the person. This ability—to be intellectually sharp and personally approachable—became a defining feature of her scientific style.

Do You Know - What is Perturbative and Nonperturbative QCD?

Perturb means to disturb slightly or to introduce a small change. In physics, Perturbation theory is a method where a complex problem is solved by starting with a simple case and adding small corrections step by step.

In Quantum chromodynamics (QCD), which describes the interactions of quarks and gluons, physicists use two main approaches:

Perturbative QCD:

When the interaction is weak (at high energies due to asymptotic freedom), calculations can be done using perturbation theory. One begins with a simple solution and improves it by adding small corrections.

Nonperturbative QCD:

At low energies, the strong force becomes so strong that it cannot be treated as a small correction. In this regime, perturbation theory breaks down. Nonperturbative QCD deals with such situations, including important phenomena like quark confinement and the formation of bound states such as protons and neutrons.

Outside formal work, she also began engaging with social issues. During her time in Mumbai, she took part in activities such as teaching women in a slum to read and write, and joining a group opposed to nuclear weapons. These experiences did not directly involve particle physics, but they deepened her awareness of inequality and responsibility—concerns that later shaped her work for women in science.

By 1982, her Visiting Fellowship at TIFR had run its course. She spent a brief period as a CSIR Research Associate at the Royal Institute of Science in Mumbai, but soon a larger professional opportunity opened up.

University of Bombay: teaching and research together

In November 1982, at around thirty years of age, Rohini joined the Department of Physics at the University of Bombay (now University of Mumbai) as a Lecturer. For the next thirteen years, until 1995, this would be her main institutional home.

The move from a focused research institute like TIFR to a large university brought changes. Teaching responsibilities increased. She now had

to prepare and deliver lectures for undergraduate and postgraduate students, set examinations and evaluate scripts, all while continuing her research. The facilities were less specialised than at TIFR, but there was a different kind of satisfaction: direct, sustained contact with students at various levels.

Former students remember her lectures as clear, demanding and energetic. She expected them to arrive prepared, but she also took the trouble to break complex topics into understandable steps, often connecting abstract ideas to simple analogies. Her time at Huzurpaga and IIT, where she herself had had to cross a language and confidence barrier, made her sensitive to students who were bright but hesitant.

Scientifically, she did not slow down. Instead, she broadened her range.

Beyond the Standard Model: supersymmetry and R-parity

By the mid - 1980s, the Standard Model had scored many successes, but it also raised new questions. Why do particles have the masses they do? What is the nature of dark matter? Is there a deeper symmetry underlying the known interactions? One attractive set of ideas was grouped under the name supersymmetry (SUSY).

In simple terms, supersymmetry proposes that for every known particle there exists a heavier “superpartner” (for example, for every fermion, a bosonic partner, and vice versa). If realised in nature, SUSY could help stabilise the Higgs boson mass, provide candidates for dark matter and unify different forces at high energies.

During her University of Bombay years, Rohini began working seriously on supersymmetric models. With Probir Roy and Xerxes Tata, she co-authored what is regarded as the first Indian paper on R-parity violation in supersymmetry.

R-parity is a quantum number assigned to particles in many SUSY models. If R-parity is conserved, the lightest supersymmetric particle (LSP) is stable and can be a dark matter candidate. If R-parity is violated, new kinds of interactions become possible, leading to distinctive decay patterns at colliders and different implications for cosmology.

By examining the consequences of R-parity violation, Rohini and her collaborators helped open up a branch of SUSY phenomenology that later attracted many younger researchers in India and abroad. Once again, her focus was characteristic: to ask how a theoretical idea would actually show itself in an experiment—through what particles, what signatures, what distributions.

Do you know? What is Supersymmetry (SUSY) and R-parity?

Supersymmetry is a theory in particle physics that proposes that every known particle has a heavier “superpartner.” It extends the Standard Model of particle physics. These partners are similar but heavier and harder to see. For example, an electron may have a partner called a *selectron*, and a quark may have a *squark*. Scientists think these partner particles might help explain big mysteries of universe, such as dark matter and how the Higgs boson behaves. Rohini Godbole, at Indian Institute of Science, studied ideas like SUSY and how to detect such particles. Her work explored how SUSY particles might be detected in experiments like those at CERN.

R-parity is a quantum number introduced in Supersymmetry (SUSY), a theory that extends the Standard Model of particle physics.

It is defined as, $R = (-1)^{3(B-L) + 2s}$, where B is baryon number, L is lepton number and s is spin. R-parity has a value of (+1) for Standard Model particles and (-1) for their superpartners. Also known as “matter-parity”. In many Supersymmetry (SUSY) models, R-parity ensures that protons – and hence all of the atoms in the universe, are unable to decay to other particles quickly by exchanging SUSY particles.

Spin, Baryon Number, Lepton Number, and R-Parity: A Simple Guide

To understand properly about R-parity, we need to understand a few “labels” or “quantities” that physicists use to keep track of particles. Think of them like an identity card for each tiny particle.

1. What is Spin?

Imagine a spinning top. It has angular momentum because it rotates. Now think about an electron. It is not really a tiny ball spinning in space, but it has a property that behaves in a similar way. This property is called spin.

Spin is a fundamental and unchangeable property of a particle, like its mass or charge.

For school students: You can think of spin as a tiny built-in property that behaves a bit like a magnetic feature. It can point in two basic ways, often called “spin up” and “spin down.” This property plays a key role in how particles behave, especially in atoms and magnetic materials.

2. What is Baryon Number?

This is like a “family number” for a group of particles called Baryons. The most familiar baryons are protons and neutrons, which sit at the centre of atoms.

The rule is simple:

- Every baryon has a baryon number of +1
- Every anti-baryon has a baryon number of -1
- All other particles (like electrons, photons, and mesons) have 0

Why is this important? Because in all ordinary physical processes we observe, the total Baryon number remains constant. It is like a cosmic accounting rule—what you start with must balance with what you end up with.

3. What is Lepton Number?

This idea is very similar, but applies to a different family of particles called Leptons. The most familiar lepton is the electron. Others include the muon, tau, and their associated neutrinos.

The rule:

- Every lepton has a lepton number of +1
- Every anti-lepton has a lepton number of -1
- All non-leptons (like quarks and photons) have 0

Again, in all ordinary situations, the total Lepton number stays constant. It helps physicists keep track of what transformations are allowed in nature.

So, analysing again what is R-Parity?

Now we come to a more advanced idea.

In theories like Supersymmetry, physicists propose that every known particle has a partner—called a “superpartner.” For example, an electron would have a partner called a selectron.

This raises an important question: If these superpartners exist, why doesn’t ordinary matter just transform into them? Why are atoms stable? This is where R-parity comes in.

The Idea Behind R-Parity

R-parity is a mathematical rule that assigns a value to particles:

- All ordinary particles $\rightarrow R = +1$
- All superpartners $\rightarrow R = -1$

But the key point is this: The total R-parity must remain the same in any interaction.

What Does This Mean?

- A single superpartner cannot appear by itself
 - Superpartners must be created or destroyed in pairs
 - The lightest superpartner cannot decay into only ordinary particles
- This last point is very important: It means the lightest superpartner would be stable.

A Simple Analogy

- Think of a school event with wristbands.
 - Ordinary particles wear green wristbands (+1)
 - Superpartners wear red wristbands (−1)
- The rule is not that they can never interact—but that the overall balance of wristbands must stay the same. So, if two red wristbands appear, they must come together. And if one red wristband remains alone, it cannot simply disappear—it has to stay.

Why is This Important?

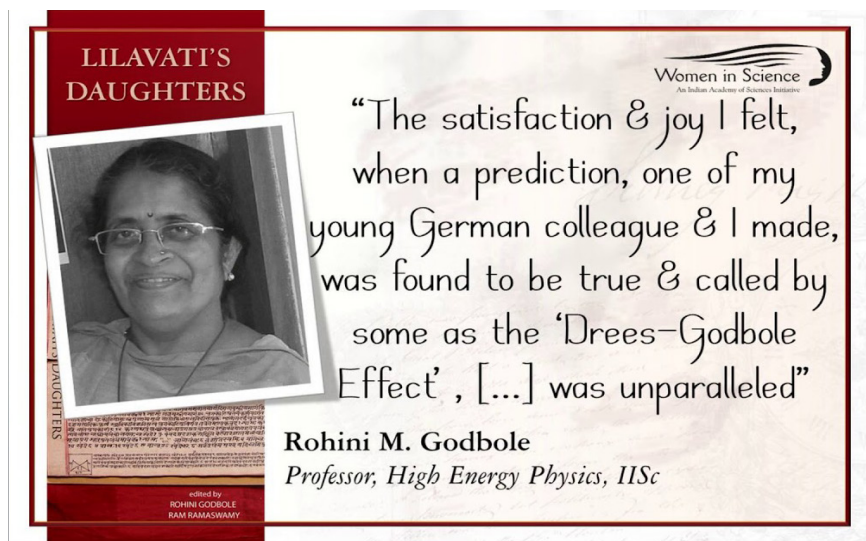
- This simple rule has a powerful consequence.
- If the lightest superpartner is stable, it could still be present in the universe today.
- Scientists believe such particles could make up Dark matter—the invisible substance that holds galaxies together.

In Short

- Spin tells us about a particle's intrinsic behaviour
- Baryon and lepton numbers act like conservation “accounting rules”
- R-parity helps keep the ordinary world stable
- And it may even point us toward understanding dark matter

This period also saw extensive collaborations with physicists abroad, including Manuel Drees, a German postdoctoral researcher she worked with during a sabbatical year at the University of Dortmund in 1988–89. Together they studied, among other things, the hadronic structure of high-energy photons—in other words, how a photon, which is fundamentally a point-like particle, can behave effectively as if it has an internal structure when probed at very high energies.

They analysed the “vacuum polarisation” around such a photon and showed that its behaviour could be described using structure functions, a tool normally used for composite particles like the proton. This work had important implications for how future colliders could be designed and used to study photon structure. It also established Rohini as an international authority in this niche but crucial area of QCD phenomenology.



(Figure 3.2: Rohini M. Godbole, *Lilavati's Daughters*)

Balancing continents: personal choices and scientific focus

During these years, science was not the only dimension of her life. Rohini spent some time in Europe as a postdoctoral researcher, and in this period, she met the German particle physicist Dr Marek Nowakowski, whom she later married. For several years, they lived a long-distance marriage: she was primarily based in India, while he worked in Europe, and they visited each other during vacations and breaks—she going abroad in summers, he coming to India around Christmas.

She later described these as some of the toughest but also most productive years of her life. The arrangement required emotional resilience and careful time management, but it allowed both partners to follow their scientific paths. She published some of her important papers during this period, proving that it was possible—though not easy—to maintain a strong research profile while managing a difficult personal life.

From Scientific Discovery to Knowledge Dissemination

Rohini Godbole's journey connects the world-class research environment of TIFR with the rich academic legacy of the University of Bombay. From advancing fundamental research at TIFR to sharing knowledge with students at University of Bombay, her path reflects dedication to science, learning, and academic excellence.

The **Tata Institute of Fundamental Research (TIFR)** is one of India's premier institutions dedicated to advanced research in fundamental sciences. Established in 1945 by the visionary physicist Homi J. Bhabha, with support from the Tata Trusts, TIFR laid the foundation for modern scientific research in India. Located in Mumbai and functioning under the Department of Atomic Energy, Government of India, the institute is internationally recognized for excellence in physics, mathematics, astronomy, biology, and computer science. Over the decades, TIFR has played a pivotal role in the development of nuclear science, space research, and advanced theoretical studies in the country. With its world-class faculty, state-of-the-art facilities, and strong global collaborations, TIFR continues to contribute significantly to scientific innovation and the advancement of knowledge.

The **University of Mumbai**, formerly known as the University of Bombay, is one of India's oldest and most prestigious public universities. Established in 1857, it has played a significant role in shaping higher education in India. The University of Mumbai offers undergraduate, postgraduate, and doctoral programs across diverse disciplines including arts, science, commerce, law, engineering, management, and technology. With hundreds of affiliated colleges across Maharashtra, it is one of the largest university systems in the world. Over the decades, it has produced distinguished alumni in politics, science, literature, business, and the arts, contributing significantly to national development and global scholarship.

Eventually, the marriage ended and they went their separate ways. Rohini did not have children, something she sometimes spoke of with honesty and without self-pity. She built a “family” of another kind: nieces and nephews she was close to, and a large community of students and younger colleagues who felt at home in her presence.

Her own experiences of negotiation and compromise, both in science and life, deepened her empathy for younger scientists, especially women, who faced pressures from family, career expectations and social norms. This empathy later became a powerful force when she began working systematically on gender and science.

A growing reputation

By the early 1990s, after a decade at the University of Bombay, Rohini had established herself as:

- A serious and reliable researcher, publishing regularly in respected international journals on topics ranging from the Standard Model to supersymmetry and collider phenomenology.
- A good teacher, who took her classroom duties seriously and guided students patiently through difficult material.
- A valued collaborator, known for her clarity, energy and ability to connect different aspects of a problem.

Her scientific output and international collaborations ensured that Indian work was visible in global discussions on particle physics. Her



(Figure 3.3: Rohini Godbole speaking at the Kaoru Hagiwara fest at the Kavli Institute for the Physics and Mathematics of the Universe in Kashiwa, Japan. Image courtesy: Tao Han, source: <https://bhavana.org.in/rohini-godbole-a-passion-for-science/>)

teaching and mentoring helped build a local community of students who could imagine themselves as part of that wider world.

Yet she felt that even more could be done if she worked in an environment where research was the central focus and where high-energy physics was being built as a specialised centre. Such an opportunity appeared in the mid-1990s, at an institution that was already a symbol of scientific excellence in India: the Indian Institute of Science (IISc), Bengaluru.

In 1995, she moved from the University of Bombay to join the Centre for Theoretical Studies (CTS) at IISc, which later became the Centre for High Energy Physics (CHEP). There, over the next three decades, she experienced the most productive period of her scientific life, wherein she guided a generation of students, and became one of the best-known faces of Indian particle physics worldwide.

The next chapter will follow her into the IISc environment: into seminars and offices where calculations were carried out late into the evening, into international committees where she spoke for India, and into classrooms where she helped students see why questions about tiny particles matter so much to our understanding of the universe.

Chapter - 04

IISc: Peak Years of Research and Mentorship (1995–2018 and beyond)

जैसा दीपें दीप लागे ।
तैसा ज्ञानें ज्ञान जागे ॥

Just as one lamp lights another,
knowledge awakens knowledge.

In 1995, after more than a decade of balancing heavy teaching with active research in Mumbai, Dr Rohini M. Godbole moved to Bengaluru to join the Centre for Theoretical Studies (CTS) at the Indian Institute of Science (IISc). A few years later, CTS evolved into the Centre for High Energy Physics (CHEP), and IISc became the institution with which her name was most closely associated. Here, over nearly three decades, she reached the peak of her scientific productivity, mentor a generation of students, and emerge as a national and international leader in particle phenomenology.

Think Like a Physicist - Try to Imagine

Suppose nature hides additional layers beyond what we know. Where would you look first — in rare events, subtle asymmetries, or unexpected energies? Designing such searches is part of a theorist's craft.

Student Reflection

As experienced scientists relocate, they transfer their expertise, methodologies, and professional ethos, demonstrating that institutions evolve not only through their structures, but through the dedication of people who shape them.

A new home for high-energy physics

IISc already had a reputation as one of India's finest research institutions, with strong departments in physics, mathematics and engineering. For Rohini, joining IISc in November 1995 meant entering a place where research was the central activity, and where she could devote more sustained time to complex theoretical problems while still teaching motivated students.

At CHEP, she found colleagues who were interested in a wide range of high-energy topics: from quantum field theory to lattice QCD, from string theory to collider physics. The culture was one of open doors and shared blackboards—a place where discussions spilled over from offices to corridors and tea tables. Rohini contributed actively to this atmosphere. She welcomed questions from students and peers, debated interpretations of data, and was not afraid to change her mind when new arguments were convincing.

Soon, she also took on leadership roles. She served as Chair of the Centre for several years and held the MSIL Chair for Physical Sciences, responsibilities that required careful balancing of administration and science. Even in these roles, she remained first and foremost a working physicist, deeply engaged with calculations, collaborations and conferences.

Rohini Godbole and the Scientific Legacy of IISc

In many ways, Rohini Godbole's journey reflects the spirit of IISc itself: deep research, intellectual rigor, global collaboration, and a commitment to nurturing the next generation of scientists.

The Indian Institute of Science (IISc) was established in 1909 by a visionary partnership between the industrialist Jamsetji Nusserwanji Tata, the Mysore royal family and the Government of India. For more than a century, IISc has become India's premier institute for advanced scientific and technological research and education.

IISc's research output is diverse, interdisciplinary and cuts across traditional boundaries. The Institute has over 42 academic departments and centres that come under six divisions.

IISc has a vibrant and diverse campus spread over 440 acres of greenery in the city of Bengaluru (formerly Bangalore), India's hub of high-tech companies, educational and research institutions, and numerous start-ups.

MSIL Chair

At IISc, MSIL Chair is one of the endowed chairs for faculty. It was established during 1985 by the Mysore Sales International Limited in the area of Physical Sciences. The Chair is for a period of 3 years from the date of award or up to superannuation, whichever is earlier.

Collider phenomenology: from photons to Higgs bosons

Scientifically, the period at IISc saw Rohini's interests deepen and widen. Some of her most influential work continued along lines she had already developed in earlier years; other projects responded to new discoveries and new machines.

High-energy photons and structure functions

Her earlier work with collaborators such as Manuel Drees on the hadronic structure of high-energy photons did not end with her move to IISc. She continued to refine and extend these analyses, showing how photons, though elementary, behave effectively like objects with structure when probed at sufficiently high energies. This has practical consequences: in electron–positron and electron–photon colliders, such structure affects cross-sections and event distributions, influencing both the design and interpretation of experiments.

The language used in these studies—structure functions, parton distributions, QCD evolution—may sound technical, but the underlying idea is simple enough for a student to grasp: when you look at an object with higher and higher “resolution” (or energy), you may begin to see features that were invisible before. Rohini's work helped physicists understand what such features would mean for photons and how to measure them.

Do You Know - Why the Higgs - Boson Matters?

The Higgs Boson is associated with a field that gives mass to many elementary particles. Its discovery confirmed an essential part of the Standard Model. Yet it also intensified the search for phenomena beyond the model, since many puzzles remain unresolved.

Preparing for the Higgs era

By the late 1990s and early 2000s, attention in particle physics was increasingly focused on the Higgs boson and on what might lie beyond the Standard Model. The Large Hadron Collider (LHC) at CERN was being constructed, and plans for various future linear colliders, such as the International Linear Collider (ILC), were under active discussion.

Rohini played a prominent role in analysing how Higgs bosons—both the “standard” Higgs and potential extended Higgs sectors in supersymmetric models—could be produced and studied at these machines. She worked on questions such as:

- How can the spin and parity of a Higgs-like particle be determined from the way it decays?
- What observables can distinguish between a Standard Model Higgs and a Higgs arising from more complex theories?
- How might supersymmetric particles show up in the data, particularly in the electroweak sector?

This required a combination of strong theoretical insight and practical sensibility. It was not enough to propose elegant formulas; one had to ask what real detectors could measure and how background processes might hide or mimic the signals of new physics.

Because of this work, she became an important voice in international discussions about future colliders. She was widely consulted on the physics potential of proposed machines, especially the ILC and its variants. Her contributions helped ensure that Indian perspectives were well represented in global planning for high-energy experiments.

Supersymmetry, dark matter and the unknown

Rohini’s interest in supersymmetry (SUSY) also continued strongly at IISc. Initially, she had explored R-parity violation and its implications; afterwards she investigated, with various collaborators, different regions of supersymmetric parameter space and their phenomenological consequences.

During studies on supersymmetry (SUSY) it was thought that one special SUSY particle, the lightest one, might explain dark matter. Dark matter is an invisible form of matter that makes up a large fraction of the universe’s mass but does not emit light. Rohini Godbole studied how such particles might be produced at colliders and worked on signals that could indicate their presence, for examples signals such as missing energy, unusual decay chains, specific angular distributions.

In 2012, scientists discovered the Higgs boson at CERN using the Large Hadron Collider (LHC), but they did not find clear evidence of SUSY yet. Instead of giving up, theorist like Rohini asked new questions:

- maybe SUSY particles are harder to find,
- maybe we need better experiments, or maybe we must improve our theories.

In public lectures, she often emphasized that this is how science truly works -

- when experiments bring unexpected results, scientists:
 - learn,
 - rethink their ideas, and
 - continue the search for truth with patience and persistence.

Peak productivity and scientific impact

One remarkable feature of Rohini's career at IISc was the timing of her peak productivity. For many scientists, publication rates and new ideas peak relatively early and then slow down as administrative duties and other responsibilities increase. In her case, colleagues have noted that her most productive period began in her mid-fifties and continued into her sixties.

Over her career, she wrote well over a hundred research papers, as well as reviews, conference reports and popular articles, and her work attracted thousands of citations worldwide. She also co-authored a substantial textbook, "Theory and Phenomenology of Sparticles", with Manuel Drees and Probir Roy, which became a widely used reference in supersymmetry. She later began work, with Sunil Mukhi, on a textbook on the Standard Model for Cambridge University Press; this project remained incomplete at the time of her passing but stands as evidence of her commitment to teaching at the highest conceptual level.

These numbers—papers, citations, books—are important indicators of impact, but they do not capture everything. Those who worked with her often emphasize her breadth: she did not focus narrowly on a single topic but contributed across a range of problems in collider physics, photon structure, Higgs studies and supersymmetry, always with an eye on how theory and experiment meet.

The mentor: building people, not just papers

If her publications made her visible in the international community, her mentoring made her unforgettable to her students. At IISc, Rohini supervised fourteen Ph.D. students, guided several M.Phil. and Master's theses,

and informally advised many more. Her students now work as faculty members and researchers in leading institutes and universities in India and abroad.

Her style as a supervisor combined high expectations with steady support. She did not treat students as assistants whose role was only to carry out tasks; she wanted them to grow into independent researchers with their own questions. In group meetings and one-to-one discussions, she would often respond to a student's answer with further questions: "Why? What does this limit mean physically? Have you checked the behaviour in that corner of parameter space?" In this way, she pushed them to move beyond mechanical calculation to genuine understanding.

At the same time, she remained approachable. Students recall that they could walk into her office with doubts—scientific or personal—and be heard with attention. She spoke openly about the challenges of research life, but she also showed, by her own example, that one could balance dedication with humour and warmth. For many, she became a long-term guide, someone they could consult even years after finishing their Ph.D.

Her mentoring was not limited to IISc. Through schools and workshops—for example, the series of Workshops on High Energy Physics Phenomenology (WHEPP), the DST-SERC Schools in Theoretical High Energy Physics (THEP), and the Asia–Europe–Pacific Schools in High Energy Physics (AEPSHEP)—she helped create spaces where young physicists from across India and the region could learn, present their work and build confidence. She was often a key organiser and always an enthusiastic participant, asking probing questions in talks and encouraging shy students to speak up.

The continuum beyond Retirement

In July 2018, Rohini reached the official age of superannuation at IISc. Formally, she "retired" as Professor of the Centre for High Energy Physics. In practice, the institute did not want to lose her, and she herself was not ready to stop. She was immediately appointed Honorary Professor at CHEP, a position she held till the end of her life.

Even as age and, later, health issues began to appear, she kept working. Papers continued to be written and submitted; talks were delivered in person or online; students and collaborators stayed in regular contact. Her last research papers appeared as late as 2024, only months before her passing, a testament to her lifelong engagement with physics.



(Figure 4.1: An undated photograph of Prof. Rohini Godbole in conversation with girl students at her cabin in CHEP, IISc, Image source: <https://rmgodbale.in/>)

During these later years, she also became an increasingly visible public voice—not only for particle physics, but for broader issues in Indian science, such as the importance of mega-science projects, the need for international collaboration, and above all, the necessity of equity and inclusion, especially for women.

A humane presence in a hard field

Particle physics is often seen as a tough, competitive field, filled with complex mathematics and large collaborations. Within this world, Rohini was known not only for her sharp mind but also for her human presence.

At conferences, she was one of the most approachable senior scientists. Younger participants quickly learnt that they could sit beside her at lunch and talk about their work; she would listen, ask questions that helped them clarify their thoughts and sometimes suggest directions or connections they had not considered. She often went out of her way to greet new faces, especially those who seemed shy or uncertain, and to introduce them to others so that they would not feel isolated.

At home in Pune during visits, and later during her final illness, she remained deeply attached to her family and old friends. Colleagues and students who met her there found the same combination of laughter, curiosity and realism that they had seen in seminar rooms and lecture halls.

Her life during the IISc years, then, cannot be separated into “science” on one side and “human” on the other. The qualities that defined her as a physicist - precision, clarity, diligence, and respect, were a direct reflection of her character as a person.

Looking ahead

By the time she completed more than two decades at IISc, Professor Rohini M. Godbole had helped shape:

- The scientific direction of high-energy phenomenology in India, through her work on top quarks, Higgs bosons, photons, supersymmetry and future colliders.
- The human landscape of Indian physics, through her supervision of students, organisation of schools and workshops, and encouragement of younger researchers.
- The institutional culture of IISc and other bodies, through her leadership positions and participation in committees.

Yet, as influential as these contributions were, another thread of her work was now drawing increasing attention: her sustained, evidence-based, and passionate engagement with the status of women in science—in India and globally. She saw clearly that scientific excellence cannot be completed if half the population faces invisible or visible barriers to participation.

Think Like a Physicist - Try to Imagine

If your teacher solves every difficulty for you, will you be ready when you have to face the unknown alone? Guidance sometimes appears in the form of further challenge.

Student Reflection

Institutions grow when individuals invest effort not only in personal success but in shared progress. Consider how cooperation strengthens knowledge.

Chapter - 05

A Voice for Women in Science

नारीशक्त्या विज्ञानदीपः प्रज्वलति ॥

The power of women ignites the
flame of scientific wisdom.

By the time Professor Rohini M. Godbole was a well-established particle physicist at IISc, she had already taken on many roles—researcher, teacher, supervisor, organiser. Yet she felt that something vital was still missing from the scientific landscape around her. She saw it in conference photographs, in faculty lists, in classroom compositions and in the quiet conversations of younger colleagues: women were present, but not in the numbers or positions that matched their talent. She grew more and more concerned with the poor representation of women in STEM subjects, especially in India.

For her, this was not merely a personal concern; it was a question about the health of science itself. If a field systematically loses capable people at different stages, can it truly claim to be searching for the best possible understanding of nature?

From individual experience to collective responsibility

Interestingly, Rohini's own early student years had not been marked by dramatic episodes of gender-based hostility. At IIT Bombay, in her M.Sc. class of seventeen students, only three were women, yet she later recalled that she did not face open discrimination from classmates or most teachers. Her own main anxiety had been about language, not about being a woman in STEM.

However, as she advanced through the system—as faculty member, senior researcher, and committee member—she observed a different pattern. Many bright women who had completed M.Sc. or even Ph.D. degrees in

science struggled to find secure positions, or left academia under pressure from family expectations, lack of support, or institutional practices that did not account for their realities. She saw talented women hesitate to apply for fellowships abroad, or to return after childbirth, or to compete for leadership roles.

Rohini's response to this was shaped by her scientific training. Instead of treating it only as a personal impression, she wanted data, analysis and structured understanding. Why were women dropping out at specific stages? Which policies helped, and which discouraged? How could institutions change, not just individuals?

Paris 2001: an international turning point

In 2001, the International Union of Pure and Applied Physics (IUPAP) organised its first International Conference on Women in Physics in Paris. India sent a delegation of women physicists, and Rohini was among them. She was invited to speak as a “success story”—a woman who had built a serious research career in physics in India.

At the conference, she listened to women from many countries—both developed and developing—describing the challenges they faced. Some stories were very different from Indian conditions; others were strikingly similar. Women from Africa and other parts of Asia told her that while they appreciated the concerns of Western colleagues, their own contexts were often closer to the Indian experience. When they heard Rohini speak, they felt that “if she can do it in those conditions, we can too.”

This experience had a deep impact on her. It made her realise that role models matter not only within a country but across regions that share social and economic realities. It also strengthened her resolve to ensure that the stories of many Indian women scientists—not just a few—would be documented and shared.

Lilavati's Daughters: telling the stories

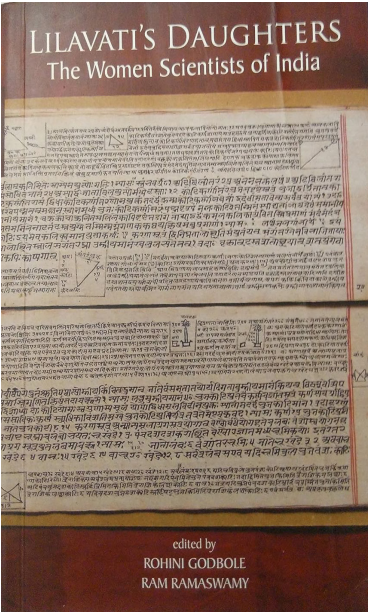
Out of these concerns was born one of her most influential projects: the book “*Lilavati's Daughters: The Women Scientists of India*”, coedited with Professor Ram Ramaswamy under the auspices of the Indian Academy of Sciences, where she later served as the founding Chair of the Panel on Women in Science.

The title paid tribute to Lilavati, the daughter of the ancient Indian mathematician Bhāskara II, who is associated—at least in popular

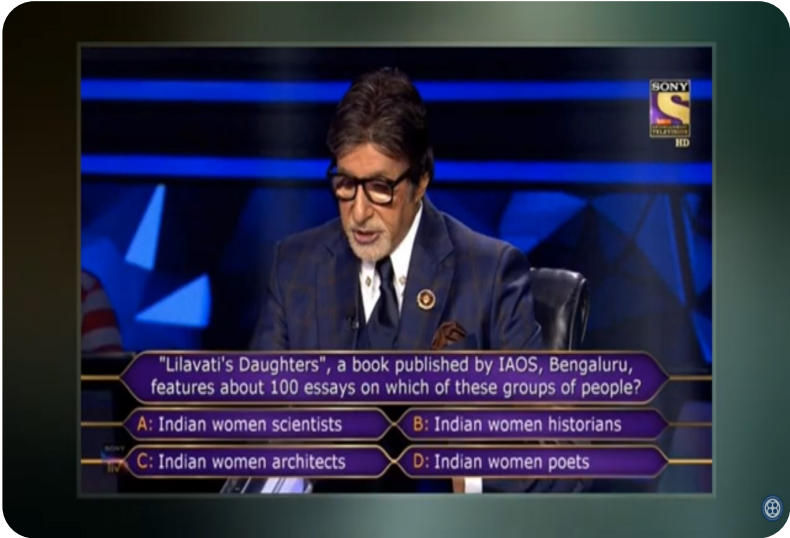
imagination—with one of the earliest mathematical texts written in a style accessible to learners. In *Lilavati's Daughters*, Rohini and her colleagues collected nearly a hundred autobiographical sketches of Indian women scientists from different disciplines and generations.

These were not idealised portraits. They were honest accounts of struggle, success, compromise and persistence: how families reacted, how institutes responded, how childcare was managed, how setbacks were handled. The book showed that Indian women had been active in science for decades, often in difficult circumstances, and that their contributions were both scientifically significant and personally courageous.

For students—especially girls—reading in schools and colleges, these



(Figure 5.1: “*Lilavati’s Daughters: The Women Scientists of India*”, edited by Prof. Rohini Godbole and Prof. Ram Ramaswami)



(Figure 5.2: The 43rd episode of 12th season of *Kaun Banega Crorepati* featured a 50-lakh worth question on *Lilavati’s Daughters* published by the IASc. Image courtesy: Internet)

stories had a simple but powerful message: you are not starting from zero. Others like you have walked this path; their experiences can help you see both the obstacles and the possibilities.

The Girl's Guide to a Life in Science: practical questions

A few years later came another important book, aimed more directly at younger readers: *"The Girl's Guide to a Life in Science"*, coedited by Rohini Godbole along with Ram Ramaswamy and Mandakini Dubey. This book combined short profiles of women scientists with explanations of different scientific fields and answers to common questions: How does one choose a specialisation? What is research actually like? Is it possible to combine family and science? What if I am the only girl in a class?

The tone was friendly but honest. The book did not promise that everything would be easy if one simply "believed in oneself". Instead, it recognised real difficulties—bias, lack of support, uneven expectations—and suggested ways to think about them, seek help and make informed decisions.

Teachers across India began using these books in workshops, seminars and classroom discussions on careers and gender. For many students, they were the first time they saw women who looked "ordinary" on the outside yet had achieved extraordinary things in laboratories, observatories, field stations and classrooms.

Panels, reports and policy conversations

Rohini's work was not limited to editing books. Within the Indian Academy of Sciences, she became the founding Chair of the Panel on Women in Science, which worked to gather data, organise meetings and prepare reports on the status of women scientists in India.

One notable output was a study titled, in essence, *"The Trained Scientific Woman Power: How We Are Losing and Why?"* This was a landmark report co-authored with Anitha Kurup, Maithreyi R., and Kantharaju B. examined how many women with scientific training were leaving the system, at what points and for what reasons. The report combined statistics with narratives, making it clear that the "leaky pipeline" (where women are lost at each successive stage) was a real, measurable phenomenon, not just a metaphor. The report powerfully refuted the simplistic idea that family responsibility was the sole reason for women leaving science, critiqued existing policies, and offered actionable items like transparency in selection and mandatory one-third representation of women on committees.

Rohini also participated in or influenced similar efforts in other bodies, such as the Indian National Science Academy (INSA), the National Academy of Sciences, India (NASI), and professional organisations like the Indian Physics Association, which eventually created a Gender in Physics Working Group. She spoke on gender and science at national and international meetings, always linking her arguments to evidence and practical suggestions rather than abstract slogans. Her approach was characteristic: firm but non confrontational, critical but constructive. She believed that institutions change best when confronted with well documented facts and clear recommendations, and when change is framed as necessary for the health of science, not as a favour to a particular group.

Guiding young minds with candour, encouragement and inspiration

Beyond committees and reports, one of the most visible aspects of Rohini's gender related work was her interaction with students, especially school and college girls. She spoke in classrooms, public lectures, television programmes and online events, often combining a short introduction to particle physics with a conversation about life in science.

Students would ask questions such as:

- “Is it harder for women to become scientists?”
- “Can you have a family and still do research?”
- “What should I do if my relatives don't take my interest in science seriously?”
- “What if I am afraid that I am not ‘good enough?’”



(Figure 5.3: Prof. Rohini Godbole and the female students of CHEP, Bengaluru 2013, Image courtesy: <https://chep.iisc.ac.in/gallery/>)

Her answers were honest. She did not say “there is no problem”; she acknowledged that biases, expectations and worries exist. But she also made it clear that these difficulties could be faced and managed, especially if one sought mentor, built peer support and chose environments where respect for work was genuine.

She often emphasised that self-doubt is normal, not a sign that one should quit. Many good scientists, she said, carry doubts throughout their careers; what matters is how they respond to them, and whether they let doubt push them to prepare better rather than to give up.

International roles and recognitions

Rohini’s work for women in science was recognised beyond India. She participated in international working groups on gender in high energy theory, contributed to newsletters and discussions hosted by organisations connected with CERN, and spoke at global events on equity and inclusion in science.

These activities were not separate from her physics; they were interwoven. She gave talks on “Doing particle physics as a woman”, on “Gender (in)equity in science: reasons and possible actions”, and on strategies for promoting gender equality in STEM. In these, she drew on her own experiences and on the collective experiences collected through panels and reports.

Her dual identity—as a respected physicist and as a thoughtful advocate for women in science—gave her words particular weight. Young researchers knew that she understood the demands of research from the inside. Senior administrators knew that she could speak their language of policy and planning. In this way, she became a bridge between individual experiences and institutional decisions.

Why this matters for students today—An inspiration for generations to come

For students reading the story of the life Rohini Godbole, her work on women in science carries several lessons:

- **Science needs diversity:** Different backgrounds bring different questions and perspectives. When large groups of people are missing from science, the field loses possible ways of thinking.
- **Fairness is a scientific value:** Just as science demands honesty with data, it should also demand honesty about how opportunities are distributed.

- **Change is possible, but it needs effort:** Committees, reports, books and talks may seem slow compared to a single dramatic discovery, but over time they change expectations, rules and habits.
- **You are not alone:** Whatever your gender or background, if you choose to pursue science seriously, you are entering a community where others have faced similar questions and have worked to make your path a little clearer.

Rohini did not solve all problems of gender inequality in science—that would be impossible for any one person. But she did something essential: she made the issue visible, discussable and actionable in spaces where it had long been ignored.

In the next chapter, we will see how the scientific and advocacy strands of her life came together in the recognitions she received—from Indian science academies, from the Government of India, from France, and from institutions like IIT Bombay and IIT Kanpur—and how she carried these honours with a sense of responsibility rather than pride alone.

Think Like a Physicist - Try to Imagine

If evidence shows that capable individuals are missing from a field, what hypothesis would you test? What mechanisms might be responsible, and how could they be corrected?

Student Reflection

When environments become more inclusive, who are the beneficiaries? Only those newly entering — or the entire community?

Chapter - 06

Honours, Recognition and Responsibility

सन्मान ही प्रेरणा । कर्तव्य ही साधना ।

सेवा हीच पूजा । यश तेथे वसे ॥

Honour is an inspiration, Duty is a sacred practice,
Service is true worship, There, success resides.

By the 2010s, Professor Rohini M. Godbole had spent nearly four decades in Indian science. Her research papers were widely cited, her students held faculty positions across the country and abroad, and her voice on women in science was heard in committee rooms, classrooms and conferences. Recognition followed naturally—but for her, awards were not endpoints. They were reminders of responsibility.

Distinguished Recipient of *Padma Shri* and the *Ordre National du Mérite*

It was, perhaps, a combination of Rohini's scientific contributions as well as her contribution to the cause of women in science that brought her two signal recognitions. In 2019, she was awarded the *Padma Shri* by the Government of India, and in 2021, the *Ordre National du Mérite* by the French Government, the latter also mentioning her important role in furthering scientific ties between India and France. Apart from these, Rohini had acquired a long list of awards and honours.



(Figure 6.1: Rohini Godbole being felicitated on her 60th birthday. Image courtesy: Sunil Mukhi, source: <https://bhavana.org.in/rohini-godbole-a-passion-for-science/>)

Recognition from scientific institutions

Scientific honours often come first from professional bodies that know a researcher's work most closely. Rohini received many such distinctions.

The Indian Academy of Sciences (IASc), where she had chaired the Panel on Women in Science, elected her as Fellow in 1992. Later, she served as Treasurer of the Academy, helping manage its finances and activities during a period of growth.

The Indian National Science Academy (INSA) awarded her the G. N. Ramachandran Memorial Lecture in 2013, an honour given to physicists of distinction. She used the lecture to discuss collider phenomenology and future experiments, linking her research to broader questions in high-energy physics.

She was elected a Fellow of the Indian National Science Academy (INSA) in 2003. She was awarded the Satyendra Nath Bose Medal by INSA in 2008.

The National Academy of Sciences, India (NASI) also honoured her work. Her election as Fellow reflected her influence across generations of researchers.

In 2010, Rohini Godbole was honoured with the C.V. Raman Mahila Vijnana Puraskara. This award was established by Swadeshi Vijnana Andolana- Karnataka (SVAK) for women scientists in the field of Physical & Mathematical Sciences.

Internationally, she was Van der Waal chair visiting Professor by the Amsterdam Institute of Physics and Astronomy, Amsterdam, 2018. She also was a member, High Energy Physics Advisory Panel, USA, 2018-2020 and also member, Board of Advisory Editors, Oxford Research Encyclopedia of Physics, 2018-2020.

Alma maters honour their own

Institutions she had once entered as a nervous student now celebrated her as an inspiration.

IIT Bombay, where she had earned her M.Sc. in 1974 and stood first, awarded her the Distinguished Alumnus Award in 2004. Returning to the campus decades later, she spoke to students about her early anxieties—language, adjustment, self-doubt—and how she overcame them. Her story showed that IIT's rigour does not exclude those who need time to find their footing.

S. P. College, Pune, her undergraduate institution, also recognised her with an alumni award. For a girl from Huzurpaga Girls' High School who had once worried about her scholarship exam, this closed a meaningful circle.

Even IIT Kanpur, which she had considered but not joined for her M.Sc., later honoured her contributions to Indian physics.

National honours: Padma Shri

The highest recognition from the Government of India came in 2019, when President Ram Nath Kovind conferred the Padma Shri upon her at Rashtrapati Bhavan. The citation read: “for distinguished service in the field of Science and Technology.”

The Padma Shri acknowledged not only her research—on top quarks, Higgs bosons, supersymmetry—but also her public outreach, her mentoring and her work for women in science. At the ceremony, she stood alongside artists, social workers and scientists, representing the diversity of Indian achievement.

Receiving such an honour from the state carried special meaning for someone whose parents had valued education as a path to dignity. She dedicated it to her family, her teachers like S. H. Patil and Bhau Sowani, and her students



(Figure 6.2: Prof. Rohini Godbole receiving Padma Shri Award from the Hon'ble President of India Shri Ramnath Kovind, Image source: <https://chep.iisc.ac.in/gallery/>)

International recognition: France’s *Ordre National du Mérite*

In 2021, France honoured her with the *Ordre National du Mérite* (National Order of Merit) in the rank of Chevalier (Knight). The award recognised her contributions to international collaboration in particle physics, particularly her work on future colliders like the International Linear Collider (ILC) and her long engagement with European physicists.

France has a strong tradition in particle physics—home to CERN’s ATLAS collaboration, the Orsay Linear Collider, and theorists like Claude Cohen-Tannoudji (Nobel laureate). Rohini’s collaborations with researchers from Dortmund, CERN, and other European centres made her a familiar figure in these networks. The award was presented by the French Ambassador to India, cementing her global standing.



(Figure 6.3: Prof. Rohini Godbole honoured with Ordre National du Mérite)



(Figure 6.4: Rohini Godbole (left) with Vaidehi Ganeshan of IGCAR, Kalpakkam, as she receives the C.V. Raman Vijnana Puraskara at the 4th National Women's Science Congress in Bangalore, on November 7, 2011. Photo Credit: Gopinathan K.)

How she carried honours

Rohini did not treat awards as personal trophies. When speaking about them, she often emphasised three things:

1. **They belong to a community:** She thanked her collaborators—D. P. Roy, Probir Roy, Manuel Drees, Xerxes Tata and many others—and her students, saying that science advances through teams, not individuals.
2. **They carry work forward:** Each recognition gave her a platform to speak about mega-science projects (like the LHC or ILC), the need for better funding for theory, and gender equity. At award lectures, she mixed physics with advocacy.
3. **They are earned through persistence:** She reminded young audiences that her path was not smooth. There were years of slow progress, limited resources, long-distance personal challenges. Honours come to those who keep working even when immediate results are uncertain.

She also used her visibility to support others. When nominating colleagues or students for awards, she wrote detailed letters highlighting their strengths. Many researchers credit her advocacy for their own recognition.



(Figure 6.5: At the Breakthrough Prize Ceremony, Geneva, 2013 where Ashoke Sen was an awardee. (L to R: Sunil Mukhi, Rohini Godbole, Sumathi Rao, and Ashoke Sen.) Image courtesy: Sunil Mukhi, source: <https://bhavana.org.in/rohini-godbole-a-passion-for-science/>)

Named lectures and professorial chairs

Beyond formal awards, Rohini was invited to deliver prestigious named lectures, a mark of respect in academia:

- J. N. Tata Lecture (TIFR)
- G. N. Ramachandran Lecture (INSA)
- Homi Bhabha Lecture
- Lectures at TIFR, IISERs, IITs and international venues.
- She held distinguished visiting positions, including the MSIL Chair at IISc (2006 - 09) and visiting professorships abroad. These allowed her to mentor students in different environments while advancing her own research.

What awards meant for Indian science?

Rohini's honours had a ripple effect:

- **For women in STEM:** Her Padma Shri and APS Fellowship showed that women could reach the top in “hard” fields like particle theory.
- **For high-energy physics:** Her recognition validated collider phenomenology as a vital area, encouraging young theorists.
- **For institutions:** IISc, IIT Bombay and others saw her success as their own, strengthening their reputation.

Most importantly, she showed that Indian scientists could contribute meaningfully to global frontiers without leaving the country permanently. Her career bridged national commitment and international excellence.

A scientist's duty

For Rohini, recognition deepened her sense of duty. Even after her 2018 superannuation from IISc, when health began to decline, she continued:

- Writing papers.
- Speaking at schools and colleges.
- Advising Ph.D. students and collaborators.
- Advocating for science policy.

Awards, for her, were not about rest. They were about passing the torch—ensuring that the next generation would ask even better questions about nature.

The final chapter will trace her last years, her passing in October 2024, and the legacy she leaves for students like you: not just papers and awards, but a way of thinking, working and caring that defines what it means to be a scientist in India today.

Think Like a Physicist - Try to Imagine

What remains after a teacher is gone? Formulas may evolve, equipment may change, but habits of honesty and persistence continue to guide inquiry.

Student Reflection

Whose encouragement has shaped your path so far? How might you extend similar support to others in the future?

Chapter -07

Legacy: A Life That Continues to Inspire

जिथे ध्येय आणि कृती एकत्र येते, ते जीवन प्रेरणादायी ठरते.

When your goals and deeds unite, your life becomes a beacon for others.



(Figure 7.1: Prof. Rohini Godbole in her office at IISc, Image courtesy: IISc Bengaluru, source: <https://bhavana.org.in/rohini-godbole-a-passion-for-science/>)

Professor Rohini Madhusudan Godbole passed away on 25 October 2024 in Pune, at the age of 71, after a period of illness borne with characteristic quiet strength. Tributes poured in from across India and



(Figure 7.2: Prof. Rohini Godbole interacting with Hon'ble Prime Minister of India Shri Narendra Modi)

the world—from students she had supervised decades ago, from collaborators in Europe and the United States, from young researchers who had read her books or heard her speak. CERN published a note of condolence. The Indian science academies issued statements. Newspapers carried front-page stories. Yet beyond these formal words, her true legacy lives in the quiet, everyday influence she continues to exert on those who knew her work.

A scientific legacy: shaping Indian particle physics

Rohini's contributions to science can be measured in several ways, each important in its own right.

- **Publications and impact:** Over her career, she authored more than 150 research papers, review articles and conference proceedings, accumulating thousands of citations. Her work on top quark signatures, photon structure functions, Higgs boson phenomenology, supersymmetry and future colliders remains referenced in planning for experiments like the LHC upgrades and the proposed ILC.

- **Textbooks and teaching materials:** The book “Theory and Phenomenology of Sparticles” (with Manuel Drees and Probir Roy) introduced a generation of students to supersymmetry. Her unfinished Standard Model textbook with Sunil Mukhi shows her commitment to making advanced physics accessible.
- **Workshops and schools:** Through WHEPP, DST-SERC Schools, and AEPSHEP, she helped train hundreds of young physicists from India, South Asia and beyond. These events created networks that endure today.

Most enduringly, her Ph.D. students now lead research groups, teach in universities and contribute to international collaborations. Through them, her way of thinking—rigorous, practical, collaborative—continues.

An institutional legacy: building cultures of excellence

Rohini helped transform institutions she touched:

- At TIFR, she was among the early phenomenologists who showed how theory could guide experiment.
- At University of Mumbai, she balanced heavy teaching with research, mentoring students in resource-constrained settings.
- At IISc CHEP, as Chair and senior faculty, she fostered a culture of open discussion and high standards.

Her leadership in academies—IASc Treasurer, INSA Distinguished Scientist, Panel on Women in Science Chair—influenced policy on funding, training and equity.

She advocated for Indian participation in mega-science projects like the LHC, AstroSat and LIGO-India, arguing that global frontiers need Indian voices.

A legacy for women and girls in science

Perhaps her most personal legacy is her work for gender equity, which changed conversations in Indian science.

Lilavati's Daughters (2008) and *The Girl's Guide to a Life in Science* (2011) gave visibility to dozens of women scientists, showing students that the path exists. These books are now in school libraries and used in career workshops.

Her reports documented the “leaky pipeline” with data, leading to practical recommendations: better childcare, transparent hiring, mentoring programmes. The Gender in Physics Working Group of the Indian Physics Association traces inspiration to her efforts.

For countless girls, she was living proof: a Marathi-medium school girl from Pune could become a Padma Shri physicist without hiding her roots or softening her questions.

Lessons for students: what you can carry forward

Rohini's life offers practical lessons and not merely abstract ideals especially for the student fraternity:

- **Curiosity can do wonders:** She was not a “child prodigy”. She worked steadily, asked questions, sought mentors like Bhau Sowani and S. H. Patil.
- **Failure helps in improvement:** Calculations failed; papers were rejected; supersymmetry searches at LHC did not find what was expected. She adjusted and continued.
- **Science is teamwork:** Her best work came from collaborations—with D. P. Roy, Probir Roy, Manuel Drees, students. No one succeeds alone.
- **Balance rigour and kindness:** She was demanding in seminars but warm in conversations; critical of weak arguments but encouraging of honest effort.
- **Think beyond yourself:** Awards like Padma Shri (2019) and Ordre National du Mérite (2021) gave her platforms to advocate for others.
- **Persist through constraints:** Limited funds, heavy teaching, long-distance marriage, health challenges—she kept contributing till the end.

A life that mattered

Rohini Godbole was not perfect. She faced personal losses, institutional frustrations, scientific disappointments. Yet she built something lasting: knowledge advanced, students empowered, conversations deepened.

In her memory, institutions established lectures and awards. IISc named a seminar room after her. Students started reading groups on her papers. Girls in Pune schools now hear her story alongside Ramanujan's or Kalpana Chawla's.

Her life reminds us: science is not just equations on blackboards. It is questions asked with courage, answers sought with patience, and a world made slightly fairer for those who follow as ‘What question will you ask next?’

Pause and Reflect: Food for Thought

Rohini's life offers questions worth thinking about quietly, discussing with friends, or writing about in your journal. Let these ideas stay with you as you make your own choices.

- **When encouragement arrives at the moment of uncertainty, it can redirect an entire life.** Rohini won a science scholarship despite no formal teaching. What does this teach us about a teacher's role?
- **Major choices often require both personal effort and collective support.** Why did she return to India after her Ph.D.? What does "responsibility" mean in science?
- **Prediction is not fortune-telling; it is disciplined imagination guided by mathematics.** How does collider phenomenology connect theory and experiment? Why does it matter?
- **When environments become more inclusive, the entire community benefits.** Her books helped girls see scientists like themselves. How can you make science welcoming?
- **Consider the balance between discovery and duty.** Should scientists work only on research, or also contribute to society?

These reflections might lead you to:

- Read the book *'Lilavati's Daughters'* to meet women scientists from different fields.
- Try sketching a particle collision showing top quark decay—imagine what detectors would see.
- Discuss with classmates: How can your school or science club make science more welcoming to everyone?

Appendix – A:

Chronological Timeline of Major Milestones

- 1952 – Born in Pune, Maharashtra.
- 1960s – Schooling marked by strong academic performance; wins state scholarship that strengthens commitment to science.
- Early 1970s – Undergraduate studies in Pune, completes B.Sc. in 1972.
- Mid 1970s – Master’s programme at Indian Institute of Technology Bombay (1974); receives highest distinction and receives the Institute Silver Medal.
- 1974 – Joins doctoral studies at Stony Brook University.
- 1979 – Completes Ph.D. in theoretical particle physics.
- Late 1970s – early 1980s – Returns to India; begins research career at Tata Institute of Fundamental Research.
- 1982 – Faculty position at University of Mumbai; active in teaching and expanding collider phenomenology.
- 1988-89 – Sabbatical year at the University of Dortmund in then-West Germany.
- 1995 – Joins Indian Institute of Science.
- 2000s – Increasing international leadership; major contributions to Higgs and beyond-Standard-Model studies.
- 2008 onward – Prominent role in national initiatives concerning women in science.
- 2012 – Higgs boson discovery energises new research directions.
- 2019 – Awarded the Padma Shri.
- 2021 – Honoured by France with the Ordre National du Mérite.
- 2024 – Passes away, leaving a vast academic and human legacy.

Appendix – B: Major Publications

A Trailblazer in Physics: Contributions and Impact

Prof. Rohini Godbole has made significant contributions to both scientific research and science communication through her publications. While no metric can fully capture the depth and quality of Prof. Rohini Godbole's work, it is remarkable to note that her scientific career spanned 48 years, during which she produced around 350 articles, interviews, reports, and conference talks, an average of seven per year, despite her many other commitments. Her work has been highly cited, with thousands of citations in international literature. The impact of her work is reflected in the more than 16,000 citations it has received, with an impressive average of about 60 citations per paper (h-index 63).

She was co-author, with Manuel Drees and Probir Roy, of the widely acclaimed textbook *Theory and Phenomenology of Sparticles*, which is a key reference in the study of supersymmetry, published by World Scientific Publishing Company.

Beyond technical physics, she has been a strong advocate for women in science, co-editing *Lilavati's Daughters: The Women Scientists of India*, a celebrated collection of biographies highlighting the achievements of nearly 100 Indian women scientists. This book was co-edited with Professor Ram Ramaswamy under the auspices of the Indian Academy of Sciences,

A more accessible version for young readers, *The Girl's Guide to a Life in Science*, was also developed under her guidance which she co-edited with Ram Ramaswamy and Mandakini Dubey.

Her final endeavour was a comprehensive textbook on the Standard Model, undertaken with co-author Sunil Mukhi under contract with Cambridge University Press. Sadly, she passed away before the work could be completed, despite the immense effort both had invested.

Appendix – C:

Further Reading and Exploration

Readers seeking to deepen their understanding of the life and profound legacy of Prof. Rohini Godbole are invited to explore the following curated selection of websites, online archives, and literary works. These resources illuminate her intellectual journey, collaborative spirit, and enduring impact on scientific research, offering both inspiration and insight for scholars and enthusiasts alike.

Works involving Rohini Godbole

- Godbole, R. M., & Ramaswamy, R. (Eds.). *Lilavati's Daughters: The Women Scientists of India*.
- Ramaswamy, R., Godbole, R. M., & Dubey, M. (Eds.). *The Girl's Guide to a Life in Science*.
- Kurup, A., Maithreyi, R., Kantharaju, B., & Godbole, R. (2010). *Trained scientific women power: How much are we losing and why?* Indian Academy of Sciences & National Institute of Advanced Studies Research Report

Articles, Lectures, and Archival Material

- Public lectures delivered at national science institutions.
- Interviews and recorded conversations on science education and gender equity.
- Rohini M Godbole: <https://rmgodbole.in/>

Publications of Rohini Godbole

- Refer to the papers at the link:
- https://rmgodbole.in/wp-content/uploads/2018/07/pub_may-2018.pdf

Background Reading for Students

- Griffiths, D. *Introduction to Elementary Particles*.
- Close, F. *The Infinity Puzzle*.
- CERN educational outreach publications.

Institutional Sources

- Materials from the Indian Institute of Science archives.
- Publications associated with the Tata Institute of Fundamental Research.

ABOUT THE BOOK

The Trailblazing Journey of Rohini Godbole chronicles the extraordinary life of a physicist who redefined Indian science. From a Pune schoolgirl winning scholarships against odds to a Stony Brook Ph.D., TIFR pioneer on top quarks, and IISc luminary in Higgs and supersymmetry phenomenology, Prof. Godbole (1952–2024) blended cutting-edge research with mentorship of research scholars. Awarded Padma Shri (2019) and *France's Ordre National du Mérite* (2021), she co-edited *Lilavati's Daughters* and *The Girl's Guide to a Life in Science*, exposing the “leaky pipeline” for women in STEM. Crafted for young ignited minds of India, this engaging life story demystifies complex scientific world while highlighting values like perseverance and inclusion. With timelines, reflections, and prompts, it is ideal for physics, career guidance, and value education. Discover how one woman's curiosity sparked a legacy that inspires India's coming generations of scientists.

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